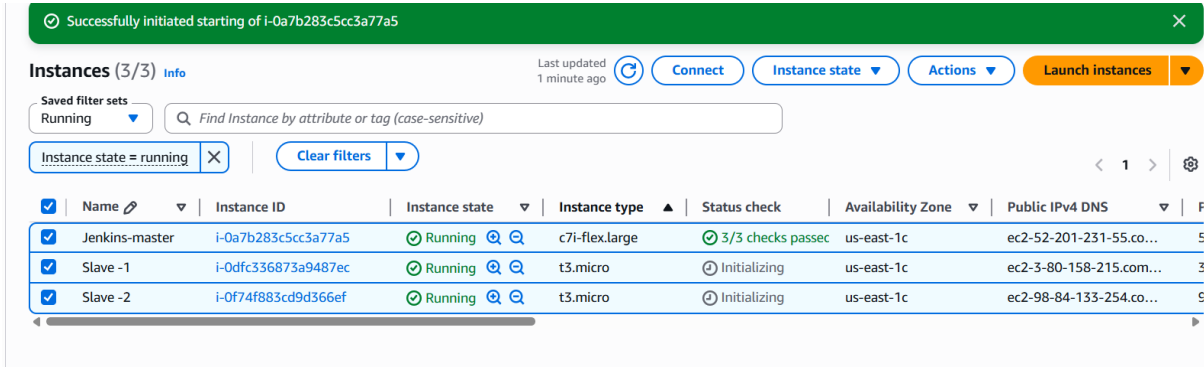


Jenkins-02

1. Configure 2 slaves machines in the Jenkins master.

- Created 2 EC2 instances for 2 slaves and for master used old one



- In slave 1 installed java 21

```
[ec2-user@ip-172-31-26-202 ~]$ sudo yum install java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 319 kB/s | 38 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1.amzn2023.0.3 amazonlinux 503 k
```

- In slave 2 installed java 21

```
[ec2-user@ip-172-31-27-165 ~]$ sudo yum install java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 320 kB/s | 38 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1.amzn2023.0.3 amazonlinux 503 k
```

- Now in Jenkins master in root added user add and passwd Jenkins

```
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-21-129 ~]$ sudo su -
[root@ip-172-31-21-129 ~]# sudo passwd jenkins
Changing password for user jenkins.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-21-129 ~]# id -u jenkins &>/dev/null || useradd -m -s /bin/bash -d /var/lib/jenkins jenkins
[root@ip-172-31-21-129 ~]# sudo passwd jenkins
Changing password for user jenkins.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

- In master created ssh keygen in su – Jenkins

```
[ec2-user@ip-172-31-26-202 ~]$ sudo su - jenkins
Last login: Wed May 13 15:54:43 UTC 2026 on pts/2
[jenkins@ip-172-31-26-202 ~]$ mkdir -p ~/.ssh
[jenkins@ip-172-31-26-202 ~]$ ssh-keyscan -H 54.89.9.228 >> ~/.ssh/known_hosts
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
[jenkins@ip-172-31-26-202 ~]$ chmod 700 ~/.ssh
[jenkins@ip-172-31-26-202 ~]$ chmod 600 ~/.ssh/known_hosts
[jenkins@ip-172-31-26-202 ~]$ cat ~/.ssh/known_hosts
|1|A2EXNKe/k1QRldp9fhqJbCocf7o=|95NlgN4gPR2Gx6dFCynw/TMeFXs= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBERYUfoscwwo
WoudbnObwAomUuz7I4XoxjBHJg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|MyUmlDrudZMInykrqg89tqv20KA=|AToxZj7IqV7dVrJh-dbt6kVNNNU= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbhxA8GZX35J7zTSASouiqws6j5S+iUBW1eNwWdJN
|1|Klu0gd/tyPrPSooq9c1aMz+61js=|ZwGIXo-x0Sfc8v0tupe2wvfer4= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBERYUfoscwwo
WoudbnObwAomUuz7I4XoxjBHJg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|TR479ACwDUXpxrncTeabf8/OqLU=|7eUeU1LEgSuxJ73dAeJd1g21xM= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbhxA8GZX35J7zTSASouiqws6j5S+iUBW1eNwWdJN
[jenkins@ip-172-31-26-202 ~]$ exit
logout
```

- In Slave 1 added the Jenkin master key given permissions

```
[root@ip-172-31-26-202 ~]# mkdir -p /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-26-202 ~]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACQ2ub0yBCTPndFN7AhoI7Sjg9fMY9jCJVzn5f4SoMMAA
VERNZ47HEeNAB0IFLkywjmVXcwTjOk47tIf25xXC7AwVORn+dZ1dYowVdcF1pjfoi01Up3vtI7H8eL8R2ScT76S0GmXgbGyyXk1eJJuQ3LXrk/xEWpP
XRCpE6ui3l0Y3s0CiJCbPnAqIknxpvZ+OwrSlo7fjHM12NhS6NhAmwr/NfhWrr+V5Ud2Ja0g6q6ebGcq2P5eQQA/wEZegmQMPBILD5tACKgTFJqDm
VbYjEATCyPdpWBk++tFuLRNVyqU8fBntNgmRDK/V/QsQATjNjSOYnArAHmfEwtuuzk0gl/fvo1NLZNcAS/+gB4t9fU5V1YrG8/E0kU6uzYTutsXm+
Tpb1fLyC9VXsDvtDnyS0uvWaKNK1nrzGmaZlUPaIKY69EwrsTl016GtrqUNf+pn2z7zSXwh2cg7gQbP8N5TmrVusTB4HsZ40N+N2Z8bg/+l0IP1sY
e1r+gBfWIdpIR7FqLU+WNpnC1n5I/ZNDrhe1hXOhUe1mPBL8M+USK+SQwVZsF9M71GjNCJTm+S6l3kwtZTUWJSDWzztNA1xj+bZ30PiSwIdF0F907A
KwBwR2qh+mBNBnhHwyMvtP+Yy1BqNP+mm+IE2NnYXknq/fcNn5HFJgDK+HgKtq5KcvtraTQ==" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-26-202 ~]# chown jenkins:jenkins /home/jenkins/.ssh/authorized_keys
chown: invalid user: 'jenkins:jenkins'
[root@ip-172-31-26-202 ~]# chown jenkins:jenkins /home/jenkins/.ssh
chown: invalid user: 'jenkins:jenkins'
[root@ip-172-31-26-202 ~]# useradd -m -s /bin/bash jenkins
useradd: warning: the home directory /home/jenkins already exists.
useradd: Not copying any file from skel directory into it.
[root@ip-172-31-26-202 ~]# chown jenkins:jenkins /home/jenkins/.ssh
[root@ip-172-31-26-202 ~]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-26-202 ~]# exit
logout
```

- Same in Slave 2 added the Jenkin master key and given permissions

```
[ec2-user@ip-172-31-27-165 ~]$ sudo su -
[root@ip-172-31-27-165 ~]# mkdir -p /home/jenkins/.ssh
[root@ip-172-31-27-165 ~]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACQ2ub0yBCTPndFN7AhoI7Sjg9fMY9jCJVzn5f4SoMMAA
VERNZ47HEeNAB0IFLkywjmVXcwTjOk47tIf25xXC7AwVORn+dZ1dYowVdcF1pjfoi01Up3vtI7H8eL8R2ScT76S0GmXgbGyyXk1eJJuQ3LXrk/xEWpP
XRCpE6ui3l0Y3s0CiJCbPnAqIknxpvZ+OwrSlo7fjHM12NhS6NhAmwr/NfhWrr+V5Ud2Ja0g6q6ebGcq2P5eQQA/wEZegmQMPBILD5tACKgTFJqDm
VbYjEATCyPdpWBk++tFuLRNVyqU8fBntNgmRDK/V/QsQATjNjSOYnArAHmfEwtuuzk0gl/fvo1NLZNcAS/+gB4t9fU5V1YrG8/E0kU6uzYTutsXm+
Tpb1fLyC9VXsDvtDnyS0uvWaKNK1nrzGmaZlUPaIKY69EwrsTl016GtrqUNf+pn2z7zSXwh2cg7gQbP8N5TmrVusTB4HsZ40N+N2Z8bg/+l0IP1sY
e1r+gBfWIdpIR7FqLU+WNpnC1n5I/ZNDrhe1hXOhUe1mPBL8M+USK+SQwVZsF9M71GjNCJTm+S6l3kwtZTUWJSDWzztNA1xj+bZ30PiSwIdF0F907A
KwBwR2qh+mBNBnhHwyMvtP+Yy1BqNP+mm+IE2NnYXknq/fcNn5HFJgDK+HgKtq5KcvtraTQ==" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-27-165 ~]# useradd -m -s /bin/bash jenkins
useradd: warning: the home directory /home/jenkins already exists.
useradd: Not copying any file from skel directory into it.
[root@ip-172-31-27-165 ~]# chown jenkins:jenkins /home/jenkins/.ssh
[root@ip-172-31-27-165 ~]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-27-165 ~]# exit
logout
```

- In master connect with ssh-keyscan of slave1 public id known hosts

```
[ec2-user@ip-172-31-21-129 ~]$ sudo su - jenkins
Last login: Wed May 13 15:14:39 UTC 2026 on pts/1
[jenkins@ip-172-31-21-129 ~]$ mkdir -p ~/.ssh
[jenkins@ip-172-31-21-129 ~]$ ssh-keyscan -H 54.89.9.228 >> ~/.ssh/known_hosts
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
# 54.89.9.228:22 SSH-2.0-OpenSSH_8.7
[jenkins@ip-172-31-21-129 ~]$ chmod 700 ~/.ssh
[jenkins@ip-172-31-21-129 ~]$ chmod 600 ~/.ssh/known_hosts
[jenkins@ip-172-31-21-129 ~]$ cat ~/.ssh/known_hosts
|1|VYuroBK00ti1od8oGSGqt/BnqmQ=|A+cAh9xnHXGQ8dnZrzCv0+Bvc40= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBERYUfoscwwo
WoudbnObwAomUuz7I4XoxjBHJg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|ErApp4kBz2BkgCrAfOT08XyRImM=|3cbni+wiJgGrd7vzwF4RumSnot4Q= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbhxA8GZX35J7zTSASouiqws6j5S+iUBW1eNwWdJN
|1|at9p4TB+B0F9ELzerJJj0YxdgFs=|jbckfInLFwb/do/z/6ZKVJgXgXI= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIokIpJnwJ2U0zD/t6qDNKuMHyvqc/IMTysd2E3R7ort
|1|TZkxthYousErw+e3U4IT7255UijM=|X8HeQUACF7JZDsA4NoTscMDm07E= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBJA7/O/DXreU
KoOMLzhKIExns9a3dUualKmpDFX28uyOCJ4VBdF00UDiyyZvhrYAKm/9RXemtrGagGisPw93dg=
|1|R6cQC1mkU9/FLJP0ddsRohb1GN4=|Zgz6Niki650eS2oUvgBmx96MXvs= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIokIpJnwJ2U0zD/t6qDNKuMHyvqc/IMTysd2E3R7ort
|1|b1T/371TjThOUmURYjTbTSYT0R8=|+0bnpk2rhM2wc4mEgfmVs0vKLeW= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBJA7/O/DXreU
KoOMLzhKIExns9a3dUualKmpDFX28uyOCJ4VBdF00UDiyyZvhrYAKm/9RXemtrGagGisPw93dg=
|1|av42F6ixs1N5QDMQ/6Mc2aHBlgE=|rS8vduHmqKpTeTpCvJzYEZYmUU= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbhxA8GZX35J7zTSASouiqws6j5S+iUBW1eNwWdJN
|1|afJempVYXU0gx+nunyXFg+nDdIw=|T5fw8ay1cPrSm54ZEThmsGkyT94= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBERYUfoscwwo
WoudbnObwAomUuz7I4XoxjBHJg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
```

- Now open Jenkins master and create new node for slave1

The screenshot shows the Jenkins 'Configure' page for a new slave node. The breadcrumb navigation is 'Jenkins / Nodes / Slave-1 / Configure'. On the left sidebar, the 'Configure' button is highlighted. Below it are links for 'Delete Agent', 'Build History', 'Load Statistics', 'Log', and 'Open Blue Ocean'. A 'Build Executor Status' box shows '(0 of 1 executor busy)'. The main configuration area contains the following fields: 'Name' (Slave-1), 'Description' (Slave-1 Machine), 'Plain text' (with a 'Preview' link), 'Number of executors' (1), and 'Remote root directory' (/home/ec2-user/jenkins/).

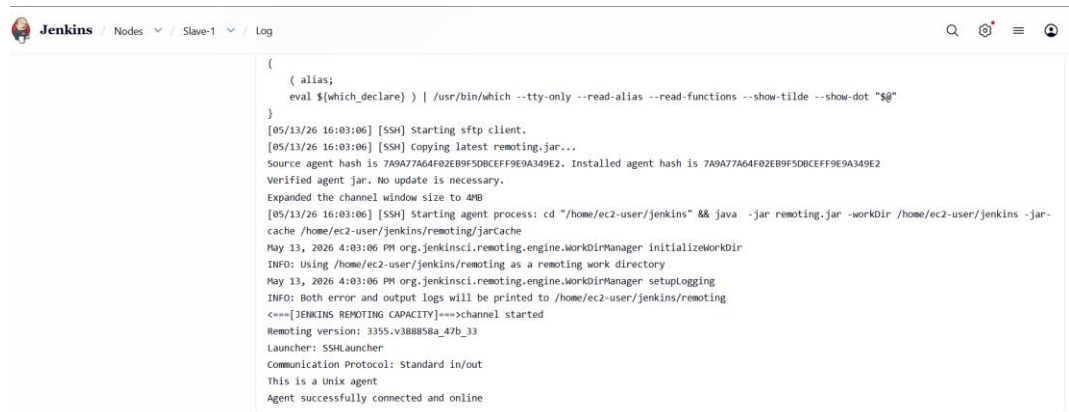
- Given label, host slave 1 public and create credentials with ec2 pem

This screenshot shows a configuration page for a Jenkins slave node. It includes the following sections: 'Labels' with the value 'linux slave1 docker'; 'Usage' set to 'Use this node as much as possible'; 'Launch method' set to 'Launch agents via SSH'; 'Host' set to '54.89.9.228'; 'Credentials' set to 'ec2-user (Slave1)' with an '+ Add' button; and 'Host Key Verification Strategy' set to 'Known hosts file Verification Strategy'.

- Save it and launch or relaunch agent check the status

The screenshot shows the 'Agent Slave-1' status page in Jenkins. The breadcrumb navigation is 'Jenkins / Nodes / Slave-1'. The left sidebar has 'Status' highlighted. The main content area shows the agent name 'Agent Slave-1', its description 'Slave-1 Machine', and a red warning icon with the message: 'Disk space is below threshold of 1.00 GiB. Only 458.37 MiB out of 458.40 MiB left on /tmp.' Below this is a 'Monitoring Data' dropdown. The 'Labels' section displays 'linux', 'slave1', and 'docker' as tags. The 'Projects tied to Slave-1' section shows 'None'. On the right side, there are buttons for 'Edit description' and 'Bring this node back online'.

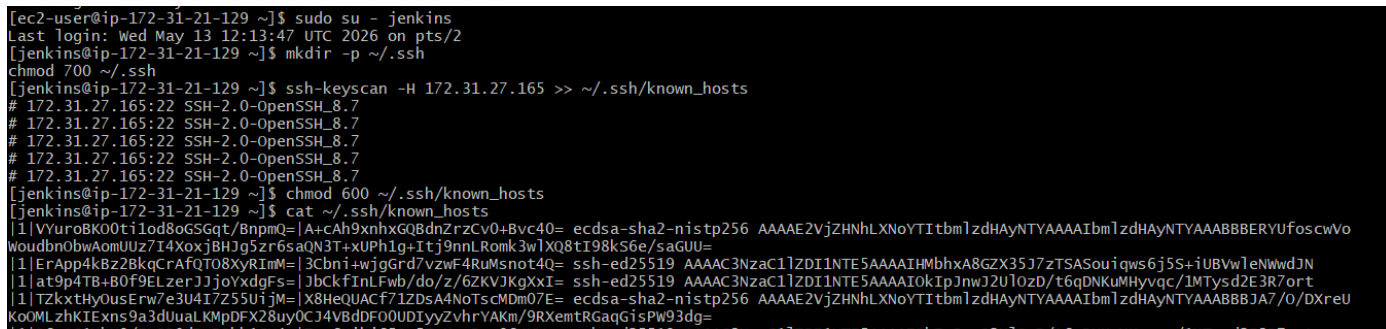
- Check logs and agent successfully connected



```
Jenkins / Nodes / Slave-1 / Log

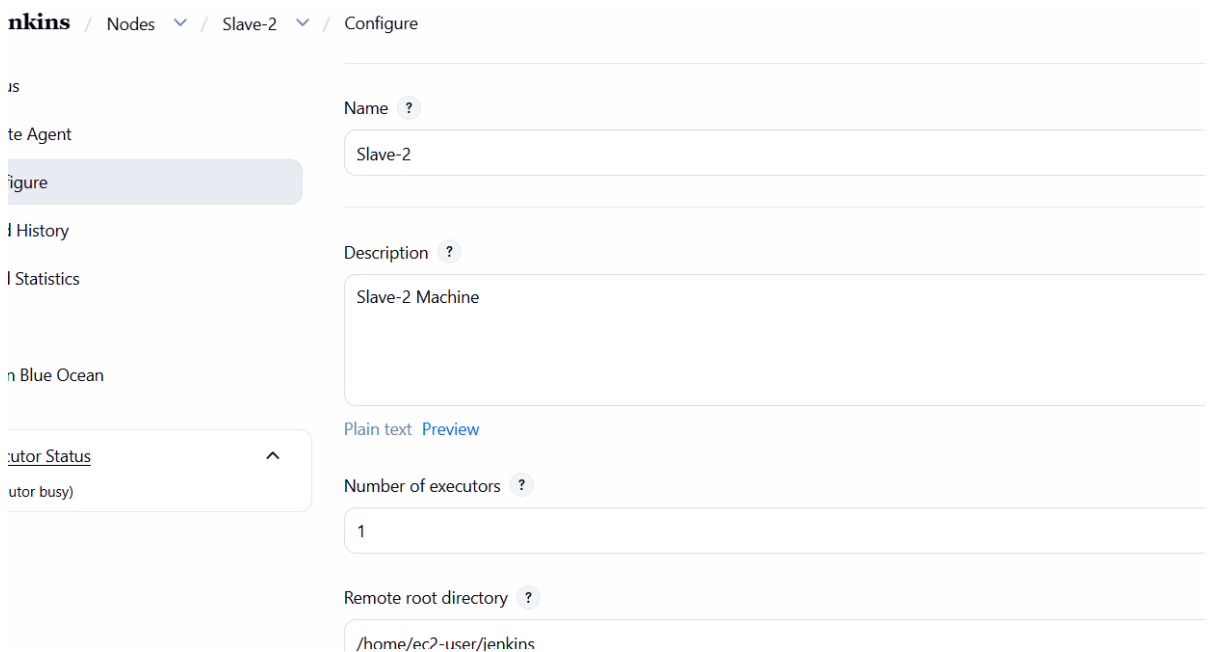
{
  ( alias;
    eval $(which declare) ) | /usr/bin/which --tty-only --read-alias --read-functions --show-tilde --show-dot "$@"
  }
}
[05/13/26 16:03:06] [SSH] Starting sftp client.
[05/13/26 16:03:06] [SSH] Copying latest remoting.jar...
Source agent hash is 7A9A77A64F02EB9F5DBCEFF9E9A349E2. Installed agent hash is 7A9A77A64F02EB9F5DBCEFF9E9A349E2
Verified agent jar. No update is necessary.
Expanded the channel window size to 4MB
[05/13/26 16:03:06] [SSH] Starting agent process: cd "/home/ec2-user/jenkins" && java -jar remoting.jar -workDir /home/ec2-user/jenkins -jar-cache /home/ec2-user/jenkins/remoting/jarCache
May 13, 2026 4:03:06 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ec2-user/jenkins/remoting as a remoting work directory
May 13, 2026 4:03:06 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ec2-user/jenkins/remoting
<===[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a unix agent
Agent successfully connected and online
```

- Now same with Slave 2 In master connect with ssh-keyscan of slave 2 private id known hosts (first logout from all roots)



```
[ec2-user@ip-172-31-21-129 ~]$ sudo su - jenkins
Last login: Wed May 13 12:13:47 UTC 2026 on pts/2
[jenkins@ip-172-31-21-129 ~]$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
[jenkins@ip-172-31-21-129 ~]$ ssh-keyscan -H 172.31.27.165 >> ~/.ssh/known_hosts
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
# 172.31.27.165:22 SSH-2.0-OpenSSH_8.7
[jenkins@ip-172-31-21-129 ~]$ chmod 600 ~/.ssh/known_hosts
[jenkins@ip-172-31-21-129 ~]$ cat ~/.ssh/known_hosts
|1|VYuroBK00tiIod8oGSGqt/BnqmQ=|A+cAh9xnhxGQ8dnZrzCv0+Bvc40= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHyANTYAAAAIbm1zdHyANTYAAABBBERYufoscwVo
WoudbnObwAomUuz7I4XoxjBHjg5zr6saQN3T+xUPh1g+Itj9nnLRomk3w1XQ8tI98kS6e/saGUU=
|1|ErApp4k8z2BkqCrAf0T08XyRiNm=|3CbnI+hwjgGrd7vzwF4RuMsnot4Q= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHMbhxA8GZX35J7zTSASouiqws6j5S+iUBVw1eNwWdJN
|1|at9p4TB+B0f9ELzerJJjoYxdgFs=|JbCkfInLFwb/do/z/6ZKVJkgXxI= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI0kIpJnwJ2U10zD/t6qDNKuMhyvqc/1Mtysd2E3R7ort
|1|TZkxthYousErw7e3U4I7Z55Uijm=|X8HeQUACf7lZDsA4NoTscMDm07E= ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHyANTYAAAAIbm1zdHyANTYAAABBBJA7/o/DXreU
KoOMLzhKIExns9a3dUuaLKmpDFX28uy0CJ4VBdF00UDIyyZvhrYAKm/9RXemtrGaQGisPW93dg=
```

- Now open Jenkins master and create new node for slave2



nkins / Nodes / Slave-2 / Configure

Name ?
Slave-2

Description ?
Slave-2 Machine

Plain text Preview

Number of executors ?
1

Remote root directory ?
/home/ec2-user/jenkins

- Given label, host slave 2 private id and create credentials with ec2 pem (Same pem key of slave 1)

Labels ?

linux docker

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

172.31.27.165

Credentials ?

ec2-user (Slave2)

+ Add

Host Key Verification Strategy ?

Known hosts file Verification Strategy

- Save it and launch the bring this node check the status

Jenkins / Nodes / Slave-2

status

Delete Agent

Configure

Build History

Load Statistics

Log

Open Blue Ocean

Executor Status ^

executor busy)

Agent Slave-2

Slave-2 Machine

Disk space is below threshold of 1.00 GiB. Only 458.24 MiB out of 458.40 MiB left on /tmp.

Monitoring Data

Labels

linux docker

Projects tied to Slave-2

None

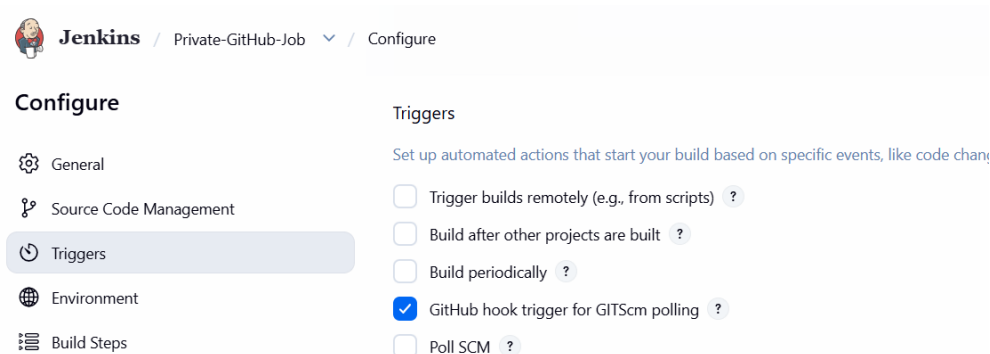
- Check the logs Successfully connected

Jenkins / Nodes / Slave-2 / Log

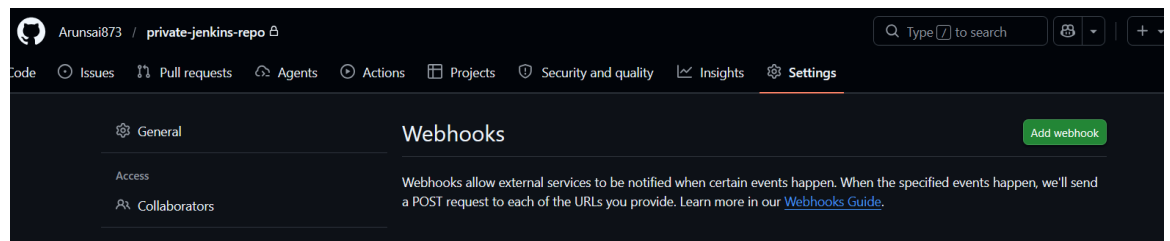
```
which ()
{
    ( alias;
    eval ${which_declare} ) | /usr/bin/which --tty-only --read-alias --read-functions --show-tilde --show-dot "$@"
}
[05/13/26 15:25:48] [SSH] Starting sftp client.
[05/13/26 15:25:48] [SSH] Copying latest remoting.jar...
[05/13/26 15:25:48] [SSH] Copied 1,407,936 bytes.
Expanded the channel window size to 4MB
[05/13/26 15:25:48] [SSH] Starting agent process: cd "/home/ec2-user/jenkins" && java -jar remoting.jar -workDir /home/ec2-user/jenkins -j.
cache /home/ec2-user/jenkins/remoting/jarCache
May 13, 2026 3:25:48 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ec2-user/jenkins/remoting as a remoting work directory
May 13, 2026 3:25:49 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ec2-user/jenkins/remoting
<===[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

2. Configure webhooks to the Jenkins job.

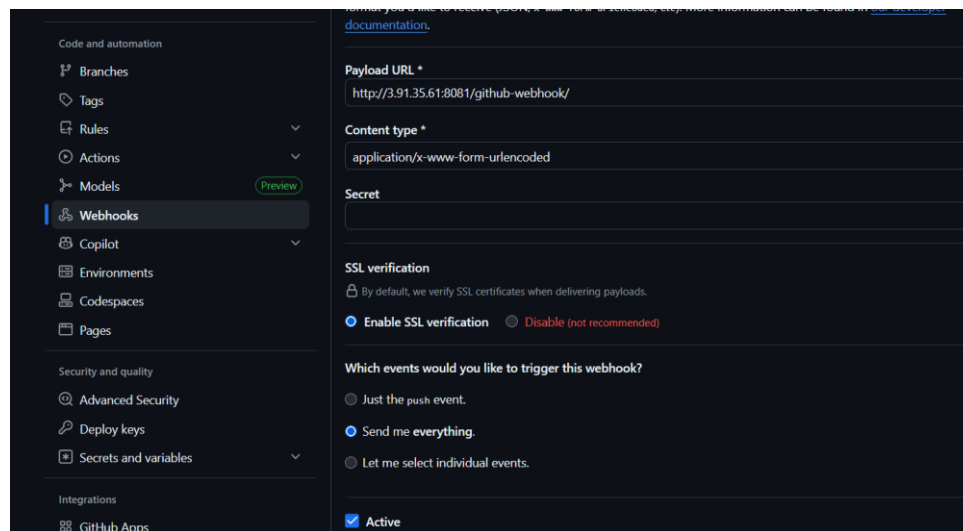
- Selected the Job which connect to GitHub repo and enable GitHub hook trigger in Build Triggers



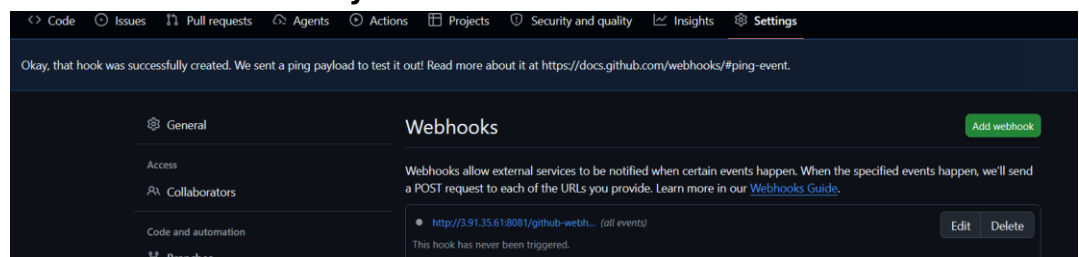
- Open GitHub and open private repo and in that settings open webhooks and add new.



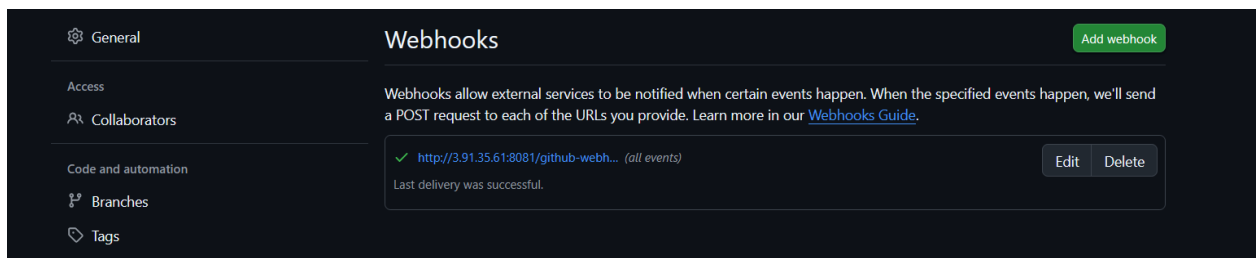
- Just post the payload URL of Jenkins and send me everything and add



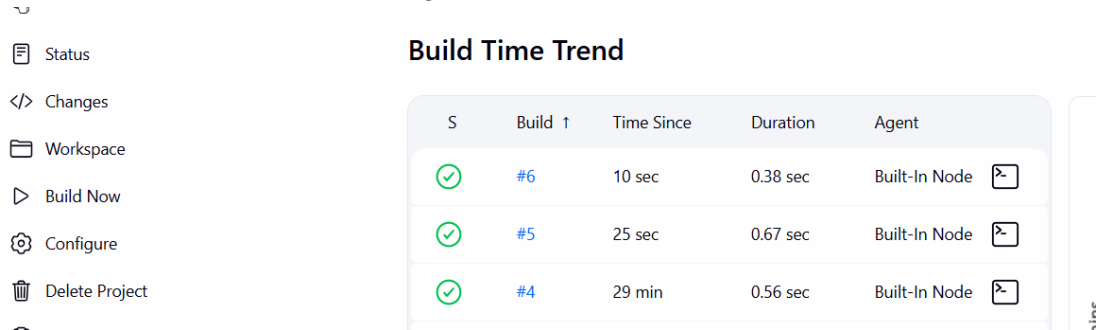
- We can see successfully webhook created



- Make any commits in repository and check in webhooks it shows

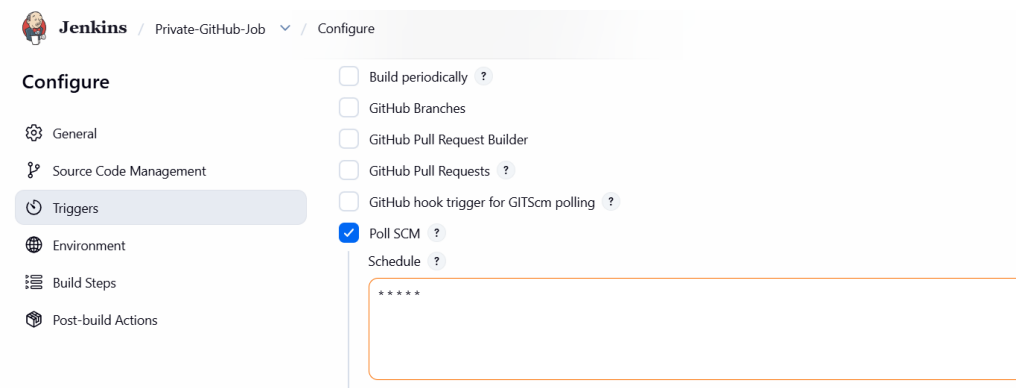


- Now we can see automatically build

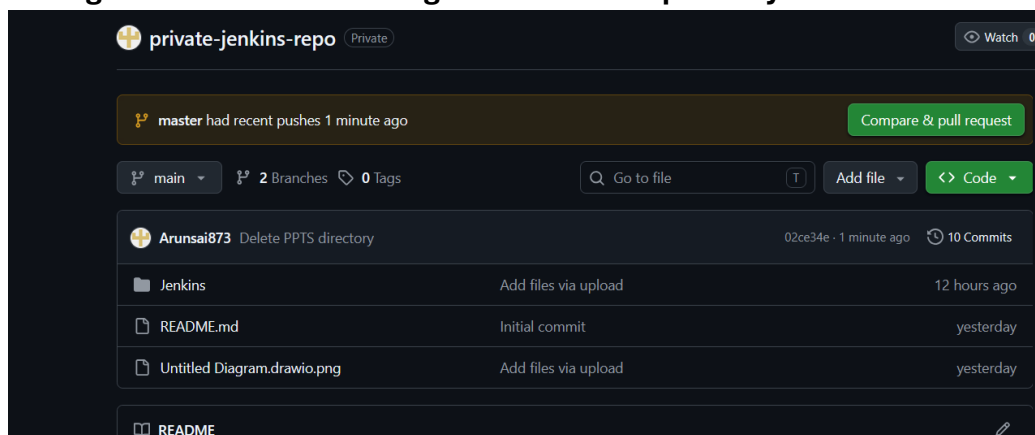


3. Configure Poll SCM and Build Periodically options in the Jenkins job.

- Open Jenkins and select the private job which is connected to GitHub and configure and enable the Poll SCM and schedule time



- Changed some commit changes in GitHub Repository and save it



- **Open Jenkins and check in build automatically start new build.**


Jenkins

/ Private-GitHub-Job
▼
/ Build Time Trend

Status

Changes

Workspace

Build Now

Configure

Delete Project

Git Polling Log

Favorite

Open Blue Ocean

Rename

Build Time Trend

S	Build	Time Since	Duration	Agent
✓	#7	1 min 8 sec	0.53 sec	Built-In Node 
✓	#6	11 hr	0.38 sec	Built-In Node 
✓	#5	11 hr	0.67 sec	Built-In Node 
✓	#4	12 hr	0.56 sec	Built-In Node 
✓	#3	1 day 7 hr	0.38 sec	Built-In Node 
✗	#2	1 day 7 hr	0.27 sec	Built-In Node 
✗	#1	1 day 7 hr	0.31 sec	Built-In Node 

Builds >



Today

 #7 00:42

Icon: S M L

mins

- **Now checked in console output also to check start by SCM**

Changes

Console Output

Edit Build Information

Delete build '#7'

Polling Log

Timings

Git Build Data

Open Blue Ocean

Previous Build

```
Started by an SCM change
Running as SYSTEM
Building on the built-in node in workspace /var/lib/jenkins/workspace/Private-GitHub-Job
The recommended git tool is: NONE
using credential private-jenkins-repo
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Private-GitHub-Job/.git # timeout=10
Fetching changes from the remote git repository
> git config remote.origin.url https://github.com/Arunsai873/private-jenkins-repo.git # timeout=10
Fetching upstream changes from https://github.com/Arunsai873/private-jenkins-repo.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
using GIT_ASKPASS to set credentials GitHub Private Repo
> git fetch --tags --force --progress -- https://github.com/Arunsai873/private-jenkins-repo.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision fa7f3f4544383b4a8beb4c4acfd6c64a9677bc8 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f fa7f3f4544383b4a8beb4c4acfd6c64a9677bc8 # timeout=10
Commit message: "Delete Servers directory"
> git rev-list --no-walk b48eddfc51b5dee2f683c3c3d10c05ae5eebf8 # timeout=10
[Private-GitHub-Job] $ /bin/sh -xe /tmp/jenkins1202854527975445316.sh
+ echo 'Private GitHub Job Success'
Private GitHub Job Success
Finished: SUCCESS
```

4. Take a backup of the Jenkins server using a bash-script.

- **Connect SSH into Jenkins EC2**



- Create Backup Directory by using the command below
sudo mkdir -p /backup/jenkins
- Now create a bash script file by using vi or vim and write script

```
ec2-user@ip-172-31-21-129:~
#!/bin/bash
DATE=$(date +%F-%H-%M)
BACKUP_DIR="/backup/jenkins"
JENKINS_HOME="/var/lib/jenkins"
tar -czvf $BACKUP_DIR/jenkins-backup-$DATE.tar.gz $JENKINS_HOME
echo "Backup Completed Successfully"
~
~
~
~
```

- Now give permissions and run it and then backup jenkins

```
ec2-user@ip-172-31-21-129 ~]$ sudo mkdir -p /backup/jenkins
ec2-user@ip-172-31-21-129 ~]$ vi backup.sh
ec2-user@ip-172-31-21-129 ~]$ chmod +x backup.sh
ec2-user@ip-172-31-21-129 ~]$ sudo ./backup.sh
tar: Removing leading '/' from member names
var/lib/jenkins/
var/lib/jenkins/.java/
var/lib/jenkins/.java/fonts/
var/lib/jenkins/.java/fonts/21.0.11/
var/lib/jenkins/.java/fonts/21.0.11/fcinfo-1-ip-172-31-21-129.ec2.internal-amzn
2023-en-.properties
var/lib/jenkins/secret.key
```

- Backup done successfully now we can verify it by ls -h

```
/var/lib/jenkins/fingerprints/77/e9/8ae60db41a583d0af6557bd02706.xml
/var/lib/jenkins/nodes/
/var/lib/jenkins/org.jenkinsci.plugins.ghprb.GhprbTrigger.xml
/var/lib/jenkins/queue.xml.bak
/var/lib/jenkins/queue.xml
Backup Completed Successfully
[ec2-user@ip-172-31-21-129 ~]$ ls -lh /backup/jenkins
total 184M
-rw-r--r--. 1 root root 184M May 13 06:23 jenkins-backup-2026-05-13-06-23.tar.gz
[ec2-user@ip-172-31-21-129 ~]$
```

5. Take a backup of Jenkins using the Thin Backup plugin.

- Open Managing Jenkins and Install Thin Backup Plugin


Jenkins / Manage Jenkins ▾ / Plugins

Plugins

Updates
 Available plugins
 Installed plugins
 Advanced settings

Install	Name ↓	Released
☐	ThinBackup 2.1.4 Miscellaneous Backups the most important global and job specific configuration files.	1 mo 1 day ago

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

ThinBackup

✓ Success

Loading plugin extensions

✓ Success

• Configure Backup Directory

ThinBackup Configuration

Backup directory ?

/backup/thinbackup

⚠ The directory does not exist, but will be created before the first run.

• Enable this backup files and save it

✓ Backup build results ?

☐ Backup build archive ?

☐ Backup only builds marked to keep ?

✓ Backup userContent folder ?

☐ Backup next build number file ?

✓ Backup plugins archives ?

• Now create directory in EC2 terminal and give permissions

```
ec2-user@ip-172-31-21-129:~  
[ec2-user@ip-172-31-21-129 ~]$ sudo mkdir -p /backup/thinbackup  
[ec2-user@ip-172-31-21-129 ~]$ sudo mkdir -p /backup/thinbackup  
[ec2-user@ip-172-31-21-129 ~]$
```

• Now open thin backup from actions and do backup now

kins ▾ / ThinBackup

ThinBackup

Settings are now integrated in global configuration.

Backup now

Restore

- **Successfully backup done**

```
[ec2-user@ip-172-31-21-129 ~]$ sudo mkdir -p /backup/thinbackup
[ec2-user@ip-172-31-21-129 ~]$ sudo mkdir -p /backup/thinbackup
[ec2-user@ip-172-31-21-129 ~]$ ls -lh /backup/thinbackup
total 0
[ec2-user@ip-172-31-21-129 ~]$
```

6. Set up a new Jenkins server and restore the backup taken in task 4.

- **Create new Ec2 instance for jenkins restore**

Instances (2) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets Choose filter set

Instance state = running Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	Jenkins-master	i-0a7b283c5cc3a77a5	Running	c7i-flex.large	3/3 checks passed	us-east-1c	ec2-34-228-43-242.co...
☐	Jenkins-Restore	i-09d32b72986331f67	Running	c7i-flex.large	3/3 checks passed	us-east-1c	ec2-34-201-69-2.comp...

- **Installed java 21 and also jenkins**

```
[ec2-user@ip-172-31-22-150 ~]$ sudo dnf install java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 288 kB/s | 38 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
```

- **Java - version**

```
Complete!
[ec2-user@ip-172-31-22-150 ~]$ java --version
openjdk 21.0.11 2026-04-21 LTS
OpenJDK Runtime Environment Corretto-21.0.11.10.1 (build 21.0.11+10-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.11.10.1 (build 21.0.11+10-LTS, mixed mode, sharing)
[ec2-user@ip-172-31-22-150 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2026-05-13 06:59:06-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://pkg.jenkins.io/rpm-stable/jenkins.repo [following]
--2026-05-13 06:59:07-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Reusing existing connection to pkg.jenkins.io:443.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.re 100%[=====>] 267 --.-KB/s in 0s
```

Jenkin installed

Package	Architecture	Version	Repository	Size
Installing: jenkins	noarch	2.555.1-1	jenkins	95
Transaction Summary				
Install 1 Package				

- Now start jenkins and check it status running or not

```
[ec2-user@ip-172-31-22-150 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-22-150 ~]$ sudo systemctl enable jenkins
[ec2-user@ip-172-31-22-150 ~]$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-05-13 07:03:12 UTC; 1min 10s ago
     Main PID: 26123 (java)
       Tasks: 37 (limit: 1014)
      Memory: 379.6M
         CPU: 20.718s
    CGroup: /system.slice/jenkins.service
            └─26123 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webro

May 13 07:03:07 ip-172-31-22-150.ec2.internal jenkins[26123]: [LF]>
May 13 07:03:07 ip-172-31-22-150.ec2.internal jenkins[26123]: [LF]> *****
May 13 07:03:07 ip-172-31-22-150.ec2.internal jenkins[26123]: [LF]> *****
May 13 07:03:07 ip-172-31-22-150.ec2.internal jenkins[26123]: [LF]> *****
May 13 07:03:12 ip-172-31-22-150.ec2.internal jenkins[26123]: 2026-05-13 07:03:12.200+0000 [id=34]
May 13 07:03:12 ip-172-31-22-150.ec2.internal jenkins[26123]: 2026-05-13 07:03:12.217+0000 [id=24]
May 13 07:03:12 ip-172-31-22-150.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuo
May 13 07:03:12 ip-172-31-22-150.ec2.internal jenkins[26123]: 2026-05-13 07:03:12.316+0000 [id=50]
May 13 07:03:12 ip-172-31-22-150.ec2.internal jenkins[26123]: 2026-05-13 07:03:12.319+0000 [id=50]
May 13 07:03:17 ip-172-31-22-150.ec2.internal jenkins[26123]: 2026-05-13 07:03:17.296+0000 [id=70]
```

- Now open in browser with public Ip address with port 8080

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

`/var/lib/jenkins/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

- Configured jenkins and ready to use

Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

Start using Jenkins

- In old Jenkins Server created manual backup

```
Last login: Wed May 13 15:12:39 2026 from 122.173.3.165
[ec2-user@ip-172-31-21-129 ~]$ ls
aws.pem  backup.sh  jenkins
[ec2-user@ip-172-31-21-129 ~]$ sudo tar -czvf /home/ec2-user/jenkins_backup.tar.gz /var/lib/jenkins
tar: Removing leading '/' from member names
/var/lib/jenkins/
/var/lib/jenkins/.java/
/var/lib/jenkins/.java/fonts/
/var/lib/jenkins/.java/fonts/21.0.11/
/var/lib/jenkins/.java/fonts/21.0.11/fcinfo-1-ip-172-31-21-129.ec2.internal-amzn-2023-en-.properties
/var/lib/jenkins/secret.key
/var/lib/jenkins/secret.key.not-so-secret
/var/lib/jenkins/plugins/
```

- Now copy to this NEW Jenkins Server

```
/var/lib/jenkins/hudson.plugins.git.GitSCM.xml
[ec2-user@ip-172-31-21-129 ~]$ scp -i aws.pem /home/ec2-user/jenkins_backup.tar.gz ec2-user@34.201.69.2:/home/ec2-user
The authenticity of host '34.201.69.2 (34.201.69.2)' can't be established.
ED25519 key fingerprint is SHA256:3Ttjx9lWTGejDb+6XLtaIAyo1o090Z4zs7b/VZeqQwM.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:3: 44.223.25.97
  ~/.ssh/known_hosts:5: 3.85.108.8
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.201.69.2' (ED25519) to the list of known hosts.
jenkins_backup.tar.gz
[ec2-user@ip-172-31-21-129 ~]$
```

- ```

#_
~\####_ Amazon Linux 2023
~\#####\
~\####|
~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~V~'-'>
~
~_._
~/_/_/_
~/m/'
Last login: Wed May 13 09:01:34 2026 from 103.24.20.200
[ec2-user@ip-172-31-22-150 ~]$ sudo systemctl stop jenkins

```

- ```
Last login: Wed May 13 09:01:34 2026 from 103.24.20.200
[ec2-user@ip-172-31-22-150 ~]$ sudo systemctl stop jenkins
[ec2-user@ip-172-31-22-150 ~]$ sudo mv /var/lib/jenkins /var/lib/jenkins-old
[ec2-user@ip-172-31-22-150 ~]$ sudo tar -xzvf /home/ec2-user/jenkins_backup.tar.gz -C /
var/lib/jenkins/
var/lib/jenkins/.java/
```

- ```
[ec2-user@ip-172-31-22-150 ~]$ sudo chown -R jenkins:jenkins /var/lib/jenkins
[ec2-user@ip-172-31-22-150 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-22-150 ~]$ |
```

- Jenkins**

🔍
⚙️
☰
👤

+ New Item
   
📋 Build History
   
🔗 Project Relationship
   
🔍 Check File Fingerprint

All

+

| S  | W  | Name               | Last Success                | Last Failure                | Last Duration | F    |
|----|----|--------------------|-----------------------------|-----------------------------|---------------|------|
| ✔️ | ☀️ | Github-01          | 2 days 1 hr <span>#1</span> | N/A                         | 3.5 sec       | ▶️ ⭐ |
| ✔️ | ☀️ | Github-02          | 2 days 0 hr <span>#1</span> | N/A                         | 0.2 sec       | ▶️ ⭐ |
| ✔️ | ☀️ | Github-03          | 2 days 0 hr <span>#1</span> | N/A                         | 0.56 sec      | ▶️ ⭐ |
| ✔️ | ☀️ | Linux-Label-Job    | 2 days 4 hr <span>#1</span> | N/A                         | 34 ms         | ▶️ ⭐ |
| ✔️ | ☀️ | Private-GitHub-Job | 16 hr <span>#7</span>       | 1 day 23 hr <span>#2</span> | 0.53 sec      | ▶️ ⭐ |
| ✔️ | ☀️ | Slave-1            | 1 hr 50 min <span>#1</span> | N/A                         | 54 ms         | ▶️ ⭐ |
| ✔️ | ☀️ | Slave-2            | 1 hr 48 min <span>#1</span> | N/A                         | 34 ms         | ▶️ ⭐ |

Build Queue

Build Executor Status

built-in node + 2 agents (0 of 0 executors busy)

Add description

- The screenshot shows the Jenkins web interface. At the top, there's a navigation bar with the Jenkins logo and name. Below it, there are tabs for 'Build Queue', 'Build History', 'Project Relationship', and 'Check File Fingerprint'. The 'Build Queue' tab is selected, displaying a table of builds waiting to be executed. The table has columns for status (S), icon (W), name, last success/failure time, duration, and actions (play and star icons). There are seven builds listed, all with a green checkmark in the 'S' column, indicating they are ready to run. The first three builds are named 'Github-01', 'Github-02', and 'Github-03'. The fourth build is 'Linux-Label-Job'. The fifth build is 'Private-GitHub-Job'. The sixth and seventh builds are 'Slave-1' and 'Slave-2'. Each build has a 'Last Success' and 'Last Failure' timestamp, a 'Duration', and two action icons (a play button and a star).

| S | W | Name               | Last Success   | Last Failure   | Last Duration | F |
|---|---|--------------------|----------------|----------------|---------------|---|
|   |   | Github-01          | 2 days 1 hr #1 | N/A            | 3.5 sec       |   |
|   |   | Github-02          | 2 days 0 hr #1 | N/A            | 0.2 sec       |   |
|   |   | Github-03          | 2 days 0 hr #1 | N/A            | 0.56 sec      |   |
|   |   | Linux-Label-Job    | 2 days 4 hr #1 | N/A            | 34 ms         |   |
|   |   | Private-GitHub-Job | 16 hr #7       | 1 day 23 hr #2 | 0.53 sec      |   |
|   |   | Slave-1            | 1 hr 49 min #1 | N/A            | 54 ms         |   |
|   |   | Slave-2            | 1 hr 48 min #1 | N/A            | 34 ms         |   |